

**Daily Goal Planning with an Artificial Intelligence (AI) Conversational Agent Enhances
Productivity and Well-being and Reduces Procrastination**

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Abstract

Self-regulation failures such as procrastination undermine goal attainment and well-being. Although structured goal planning can help bridge the intention–action gap, delivering such guided self-regulatory support at scale remains challenging. Generative AI chatbots offer adaptive, conversational assistance that may provide scalable guidance, yet it remains unclear whether such adaptive dialogue meaningfully enhances goal pursuit and well-being outcomes. The current pre-registered experiment examines whether delivering a structured daily goal-planning intervention through a generative AI chatbot improves goal pursuit and well-being outcomes relative to a rule-based, non-AI control. Compared to the control condition, the AI chatbot condition showed greater pre-test to post-test reductions in procrastination, and greater increases in implementation intentions and productivity. Exploratory analyses further demonstrated that the AI chatbot condition had greater improvements in affective well-being and reductions in mind wandering. Moderation analyses indicated that effects were largely consistent across levels of goal commitment, trust in AI, and attitudes toward AI. These findings highlight the potential of AI conversational agents as a scalable tool for supporting everyday goal pursuit and well-being.

Keywords: Self-regulation, Goal Pursuit, Procrastination, Productivity, Artificial Intelligence

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Achieving personal goals and staying productive can be challenging due to self-regulation failures such as procrastination, premature goal disengagement, and failure to act on intentions (Gollwitzer & Sheeran, 2006; Steel, 2007; Van Eerde, 2000). These challenges highlight the central role of self-regulatory capacity in effective goal pursuit. Individuals with strong self-regulation capacities tend to exhibit better psychosocial functioning, improved academic competence, and a stronger goal pursuit (Billore et al., 2023; Distefano et al., 2021; Miller et al., 2015), whereas deficits in self-regulation are associated with poorer academic performance, behavioural dysregulation and goal disengagement (Billore et al., 2023; Cambron et al., 2017). Among the self-regulatory challenges, procrastination is particularly prevalent, with more than 80 per cent of university students reporting engaging in it (Fentaw et al., 2022; Mahasneh et al., 2016). This voluntary delay of intended tasks (Klingsieck, 2013; Steel, 2007) is consistently associated with adverse psychological outcomes, such as depression, anxiety and stress (Johansson et al., 2023; Rozental et al., 2022), alongside unhealthy lifestyle behaviours, including poor sleep quality and increased physical inactivity (Johansson et al., 2023; Sirois & Pychyl, 2016). At a broader level, these self-regulation lapses illustrate a mismatch between intention and action (Steel & Klingsieck, 2016; Steel, 2007; Sheeran, 2002), which derails individuals from pursuing their goals due to distractions, temptations, and disruptive thoughts and feelings (Sheeran & Webb, 2016). This recurring gap underscores the need for strategies and tools that help individuals translate their goals and intentions into consistent action.

Research indicates that there are several evidence-based strategies for effective goal pursuit. One widely supported effective strategy is the formation of implementation intentions (Wieber et al., 2015). Implementation intentions are considered synonymous with

action planning, which involves the formation process of when, where and how to act, linking goal-directed behaviours to environmental cues (Gollwitzer, 1999; Sniehotta et al., 2005).

Another self-regulation strategy is coping planning, which is barrier-focused and helps individuals overcome obstacles by establishing a mental link between anticipated risk situations and suitable coping resources, often in an if-then format (Sniehotta et al., 2005).

Research shows that forming implementation intentions is effective in promoting goal striving and goal-directed response, with meta-analyses finding a medium-to-large improvement in goal attainment (Gollwitzer & Sheeran, 2006). More generally, time management behaviours and weekly planning behaviour, such as task prioritisation, are linked to greater perceived control of time, better performance and well-being outcomes (Aeon et al., 2021; Claessens et al., 2007; Uhlig et al., 2023). These findings provide a strong empirical foundation for how self-regulation mechanisms, from specific if-then to planning habits, can help bridge the gap between intention and behaviours, thereby strengthening goal pursuit and counteracting procrastination.

Although implementation intentions are powerful, their benefits can be difficult to sustain when individuals rely entirely on self-administered planning. Planning interventions are often delivered in individual sessions with structured planning components, such as lectures, workbooks and online videos, and self-monitoring of behaviours (Ghisi et al., 2015; Lin et al., 2022). The cues for action from self-guided plans may be poorly specified by users, whereas implementation intentions can still be effective even with minimal user information (Armitage, 2009; Hagger & Luszczynska, 2014). In the absence of ongoing, personalised feedback, the enactment of plans may be attenuated, as behavioural change is stronger when support is tailored and delivered more frequently (Fry & Neff, 2009; Pirolli et al., 2017; Noar et al., 2011). Structured coaching processes that involve goal clarification, action planning, and iterative refinement based on feedback are associated with greater improvements in

coping and goal-directed self-regulation across domains (Chawla et al., 2016; Grant, 2013; Theeboom et al., 2014; Jones et al., 2016). However, traditional human coaching is resource-intensive and not always accessible, which limits the scalability of guided goal support. Consequently, there is a need for delivery formats that retain the benefits of guided, interactive planning while remaining accessible and scalable.

Recent advances in Artificial Intelligence (AI) have made conversational chatbots a promising avenue to meet this need by delivering personalised coaching interventions at scale. Modern generative AI models, such as ChatGPT, leverage natural language processing to interpret open-ended human input and engage in human-like conversations, providing users with personalised feedback and structured coaching guidance (Bhattacharjee et al., 2023; Khanna et al., 2015; Yigci et al., 2025). Unlike earlier rule-based chatbots that lacked context awareness and gave scripted responses (Al-Amin et al., 2024; Luo et al., 2021), modern conversational chatbots enable natural interactions, rated higher in empathy, engagement and usefulness, and are capable of providing companionship and coaching (Alazraki et al., 2021; Brenda, 2024). AI chatbot-delivered coaching is scalable as it can serve many users simultaneously, providing cost-effective, 24/7 access without breaks while leveraging vast data to deliver evidence-based interventions (Passmore & Tee, 2023). There is also emerging evidence that suggests that AI-integrated interventions contributed to improvements in well-being (Alabed et al., 2024; Hu et al., 2025; Hung et al., 2025; Inkster et al., 2018). AI chatbots developed with an evidence-based foundation were comparably effective to human coaches in improving goal attainment (Terblanche et al., 2022). These findings highlight the potential of AI chatbots to serve as effective and engaging goal-support tools.

The effectiveness of conversational AI chatbots may stem from their ability to evoke interpersonal processes such as working alliance and trust. Although these agents are not

human, conversational chatbots foster a sense of digital working alliance in users, one that mirrors the therapeutic bond found in human relationships (Heim et al., 2018; Mai et al., 2022). Stronger perceived alliances with digital coaches are associated with greater credibility of intervention and higher expectations of treatment success (Heim et al., 2018). Emotional trust, the willingness to be vulnerable and act on AI-generated suggestions, is also a key predictor of adherence (Glikson & Woolley, 2020). These findings suggest that the effectiveness of AI-based interventions may depend on their ability to evoke relational processes such as working alliance and trust. In the context of goal pursuit, AI-based interventions that foster perceived alliance and emotional trust may strengthen motivation and sustained engagement with planned actions.

Building on self-regulation strategies and the potential for delivering them with AI-generated feedback, the current study translates this evidence base into a structured, daily goal planning sequence delivered through a chatbot. Accordingly, the chatbot guides participants in working toward their own overarching weekly goal. The daily goal planning process begins by assessing participants' current progress against their goal, supporting self-monitoring and discrepancy reduction by comparing one's current state to their self-set goals that serve as a reference value (Carver, 2018; Thürmer et al., 2020). Participants then externalise all anticipated tasks alongside their available time in a day, which supports realistic planning by making it easier to match tasks realistically to the time they have (Grinschgl et al., 2021). Tasks are subsequently broken down into concrete, actionable steps and prioritised, a process shown to reduce ambiguity and choice overload (Chernev et al., 2015; Locke & Latham, 2002). Timeboxing is then applied to the first step prioritised, transforming intentions into a specific commitment regarding when, where, and for how long action will occur, thereby forming an association between mental representations of situational cues and goal-directed behaviour (Gollwitzer, 1999; Sheeran et al., 2006).

Participants subsequently formulate if-then plans for anticipated obstacles, which support persistence by pre-specifying coping responses when barriers arise (Riddell et al., 2022; Wieber et al., 2015). Finally, participants report completion of the first step, providing a mechanism for progress monitoring that has been shown to promote goal attainment (Harkin et al., 2016).

The present study investigates whether delivering this structured goal planning intervention through a generative AI chatbot enhances goal pursuit outcomes more effectively than a non-AI scripted chatbot control. Both the AI and control conditions deliver the same structured daily goal planning content. However, only the AI-driven chatbot provided context-aware dialogue with tailored feedback and encouragement, whereas the control condition relied on pre-written scripts, enabling a direct comparison between the delivery of the intervention. We hypothesised that participants receiving the daily goal planning coaching from a generative AI chatbot would show greater improvements in their productivity, goal progress, goal realisation, implementation intentions, action and coping planning, working alliance, along with larger reductions in procrastination, compared to those receiving the scripted coaching. Furthermore, we explored several secondary outcomes such as cognitive failures, mind wandering, life satisfaction, positive affect, and negative affect. The AI agent's personalised, context-aware responses were expected to bridge the intention-action gap, and such AI-assisted daily goal planning may offer an accessible and effective approach for enhancing goal enactment.

Transparency and Openness

The current study's design and analysis plan were pre-registered (<https://aspredicted.org/wktr-vm5n.pdf>). The pre-registration, along with all relevant materials and data, have been made publicly available on ResearchBox (#6099; http://researchbox.org/6099&PEER_REVIEW_passcode=XNDZZQ).

Method

Participants

A total of 143 undergraduate students from a local university in Singapore were recruited through the institution's subject pool system. Participants received two course credits for completing the study, which consisted of two surveys with five daily tasks. In accordance with the exclusion criteria, participants who did not complete both surveys were excluded. An additional three participants were excluded for failing to complete the five daily tasks as instructed, resulting in a final sample of 137 participants (AI Condition: $n = 71$; Control: $n = 66$). A priori analysis using G*Power 3.1.9 (Faul et al., 2007) indicated that a sample size of 126 participants, with 63 per group, was required to detect a medium-sized Group $\times$ Time interaction (Cohen's $f = 0.25$, equivalent to partial $\eta^2 \approx .06$) in a 2×2 mixed ANOVA with 80% power. The final sample size was slightly larger than the preregistered target sample of 126 participants due to higher than expected retention. Data collection was approved by a local Institutional Review Board [IRB-25-055-A044-M1(825)].

Table 1

Demographics of Participants recorded at Pre-Test

Characteristic	<i>M (SD) or %</i>	Observed Range
Sex (% Female)	80.29 %	-
Age (years)	21.03 (1.75)	18–27
Household Income ¹	3.66 (1.64)	1–6
Ethnicity (% Chinese)	86.86 %	-

Note. ¹ Monthly household income was measured using a 6-point scale (1 = $< \$2000$, 2 = $\$2000$ – $\$5999$, 3 = $\$6000$ – $\$9999$, 4 = $\$10,000$ – $\$14,999$, 5 = $\$15,000$ – $\$19,999$, 6 = $> \$20,000$)

Procedure

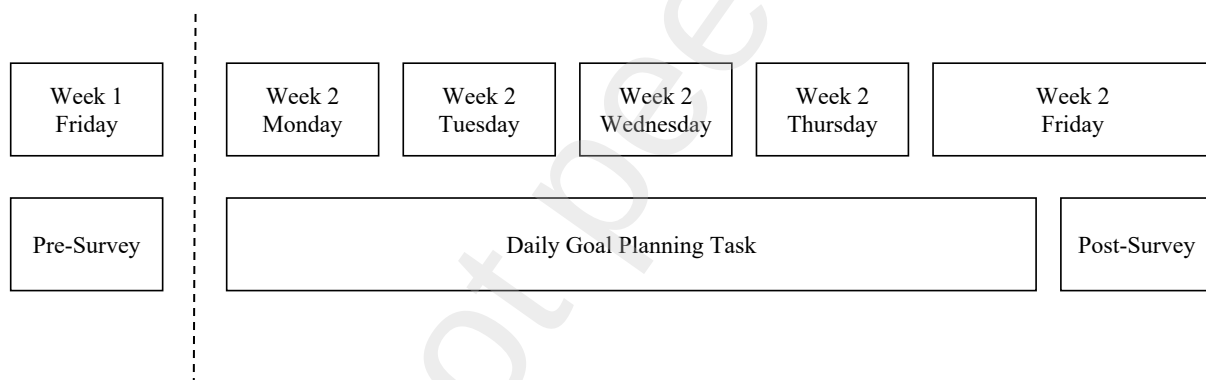
This experimental study employed a 2 (Condition: Experimental AI chatbot vs. Control non-AI chatbot) $\times$ 2 (Time: Pre-test vs. Post-test) mixed design. Upon signing up for the study, participants were informed that the study examined how completing daily tasks affected mental energy—a cover story to conceal the actual research objectives and minimise participants' expectations in relation to the assigned condition. After providing informed consent, participants completed a pre-test survey on Qualtrics, in which they set an overall goal and deliverable to be completed by the following Friday. Participants were then randomly assigned to record their goal and deliverable using either the AI or control chatbot, and were instructed to return to the survey to complete the rest of the outcome measures and demographic information. They were instructed not to begin their daily tasks until the following Monday.

During the subsequent week, participants completed five daily chatbot tasks, one task during each weekday, using their assigned chatbot. In both conditions, participants engaged in daily goal planning through an intervention delivered via a Telegram chatbot interface. Each day, they initiated the task by tapping a “Start Today’s Task” button on the chatbot, mirroring real-world self-regulation in which individuals decide when to plan rather than being prompted at a fixed time (Figure A1, Figure A2). Both chatbots guided participants through the same sequence of goal planning steps: assessing current progress, listing all tasks and available time, breaking tasks into concrete actions and prioritising them, timeboxing the first step, and formulating if-then plans for potential obstacles. This intervention was designed to help participants translate their weekly goal into concrete daily actions. A daily task was considered complete when participants tapped the “Done” button to indicate completion of their planned first step. Reminders were sent through the chatbot at noon to begin their daily task, followed by a second reminder at night for those who had not yet completed their daily tasks for the day.

The two conditions differed only in the nature of the chatbot interaction. In the control condition, participants interacted with a scripted non-AI Telegram chatbot. While in the AI condition, participants used an AI-powered chatbot that generated dynamic, conversational responses while following the same intervention sequence (see Table 2 for a detailed comparison). After verifying that participants finished all five daily tasks, they received an email with instructions to complete the post-test survey on Qualtrics that measures the post-intervention outcomes and an online debriefing that revealed the full purpose of the study and participants' assigned condition. The overall study flow is presented in Figure 1.

Figure 1

Procedure of Study



Control Condition (Non-AI Rule-Based Scripted Chatbot)

Participants assigned to the control condition interacted with a scripted, non-AI Telegram chatbot that delivered a one-way goal planning structure. The system functioned similarly to a structured journaling tool, presenting fixed prompts in the same sequence each day. Responses were prewritten and did not adapt to participants' input, instead guiding users through the goal planning process in a standardised manner. The chatbot progressed through each prompt regardless of participants' responses and maintained a neutral tone throughout. It did not provide personalised suggestions, adjust wording based on individual input, or store any information within or across sessions. As a result, participants completed the goal

planning process independently, using the chatbot as a structured, one-way journaling interface (see Figure A1).

Experimental Condition (Adaptive AI Dynamic Chatbot)

Participants assigned to the experimental AI condition interacted with an adaptive, conversational goal planning chatbot, powered by the OpenAI GPT-4o-mini model. This lightweight model was selected to enable fast, real-time text responses (OpenAI, 2024), and the system prompt further constrained responses to approximately 120 words to enhance clarity. The chatbot functioned as a goal planning coach that facilitated self-regulatory dialogue while following the same structural steps as the control version. Unlike the scripted control chatbot, the AI system generated dynamic, context-aware responses to provide individualised support throughout the daily planning intervention. To ensure continuity across sessions, the chatbot stored participants' weekly goals in its internal memory, allowing it to reference these goals during subsequent daily interactions. Within each session, it also maintained short-term memory of prior exchanges, enabling it to request clarification when responses were vague and to provide feedback tailored to participants' stated plans and constraints. The chatbot's tone was configured to be collaborative, supportive, and non-judgmental while remaining natural and conversational. For example, when available, it addressed participants by their first name and kept messages concise and goal-focused. After participants indicated completion of the first step, the chatbot offered brief encouragement and light productivity-oriented support for reflection. However, it did not restart the structured planning sequence or request another completion of the first step until the participant initiated the next day's task (see Figure A2).

Table 2

Daily Goal Planning Intervention Step Across Conditions

Step	Control Condition	AI Condition
Assessing current progress	What is your current progress toward your goal so far?	Open with encouragement referencing stored weekly goal, contextualise as a fresh start (Monday) or acknowledges prior completion (Tuesday to Friday).
Listing all tasks and time availability	List the tasks you plan to do today to work toward your overall goal. How many focused hours do you realistically have today?	Request for incomplete responses, if any.
Breaking tasks into concrete actions and prioritisation	Please break your tasks into small, concrete steps if needed, and think about why each task is important. Rearrange them in order of priority, starting with the task to be done first.	Guides further breakdown into concrete steps by suggesting up to two micro-steps.
Timeboxing for their first step	When will you start the first task today? Where will you work on it, and how long will you work on it?	Suggestions include time references and are aligned with the Asia/Singapore timezone when specific timings are mentioned.
Formulating if-then plans for obstacles	What will hold you back from starting this first step? And what would you do if it happens? (Complete this sentence: If [that obstacle happens], then I will [quick action to fix it].)	Support participants if-then plans with relevant suggestions or supportive framing.
Accountability & Task Completion	Check back in when you've finished the first step by sending 'Done' and I'll send you your exit ticket for today.	Confirms the plan and encourages participants to come back after finishing the first step. Enters wait mode until participants replies 'Done'.

Materials

Measures of Main Outcomes

Productivity. Weekly productivity was measured using the 3-item Weekly Productivity Scale. Participants rated their agreement on a scale from 1 (*Strongly disagree*) to 5 (*Strongly agree*) for items such as “In the past 5 days, I was productive.” The scores were averaged across all items after reverse scoring, with higher scores indicating greater

productivity. Internal consistency was acceptable at both pre-test ($\alpha_{pre} = .697$) and post-test ($\alpha_{post} = .695$).

Procrastination. Weekly procrastination was measured using the 6-item Weekly Procrastination Scale (Steel, 2002). Participants rated their agreement on a scale from 1 (*Very seldom or not true of me*) to 5 (*Very often true, or true with me*) for items such as “I procrastinated in the past 5 days.” The scores were averaged across all items with higher scores indicating greater procrastination. Internal consistency was good across pre-test and post-test for procrastination ($\alpha_{pre} = .902$; $\alpha_{post} = .893$).

Weekly Implementation Intention and Goal Progress. Using the Weekly Implementation Intention and Goal Progress Scale, implementation intention was measured with a single item (“In the past 5 days, I have taken steps to achieve my goals for the project I’ve chosen”), and goal progress was measured with a separate single item (“In the past 5 days, I have seen progress towards my goals for the project I’ve chosen”). Participants rated their agreement, using a 5-point scale from 1 (*Strongly disagree*) to 5 (*Strongly agree*). Higher scores indicated greater implementation intention or goal progress.

Goal Realisation. Goal Realisation was measured using the 4-item Goal Realisation Scale (Łaguna et al., 2023), which includes items such as “I have achieved my goal that I planned to achieve for the past 5 days.” Participants indicated their agreement on a scale from 1 (*Strongly disagree*) to 7 (*Strongly agree*). Scores were averaged with higher scores reflecting greater goal realisation. Internal consistency was high at both pre-test and post-test ($\alpha_{pre} = .877$; $\alpha_{post} = .848$).

Action and Coping Planning. Action and Coping Planning were measured using a 6-item scale adapted from Sniehotta et al. (2005). The scale included two 3-item subscales: (a) action planning (e.g., “I have made a detailed plan for when I will work on this goal”) and (b) coping planning (e.g., “I have made a detailed plan what to do if something interferes with

my goal”). Participants rated how true each statement was for them for the past 5 days on a 5-point scale ranging from 1 (*Not at all true*) to 5 (*Completely true*). Scores were averaged separately for each subscale, with higher scores indicating greater action or coping planning. Internal consistency was good at both pre-test and post-test for action planning ($\alpha_{pre} = .827$; $\alpha_{post} = .899$) and coping planning ($\alpha_{pre} = .901$; $\alpha_{post} = .940$).

Working Alliance. Working alliance was measured using the Digital Working Alliance Inventory (DWAI, Goldberg et al., 2022). Participants rated their agreement with each item on a scale from 1 (*Strongly agree*) to 7 (*Strongly disagree*). At pre-test, participants rated their initial interaction with the chatbot that day. At post-test, participants rated their experience over the past 5 days. The scale includes three 2-item subscales measuring (a) goal (e.g., “I trust the chatbot to guide me towards my personal goals”), (b) task (e.g., “The chatbot encourages me to accomplish tasks and make progress”), and (c) bond (e.g., “The chatbot supports me to overcome challenges”). The overall working alliance score was calculated by reverse-coding all six items and then averaging them, with higher scores indicating a stronger working alliance. Internal consistency was good at both pre-test and post-test ($\alpha_{pre} = .874$; $\alpha_{post} = .910$).

Moderator Outcomes

Goal Commitment. Goal commitment was measured using the 9-item Goal Commitment Measure (Hollenbeck et al., 1989). Participants rated their agreement with statements such as “I am strongly committed to pursuing this goal” on a scale from 1 (*Strongly disagree*) to 5 (*Strongly agree*). The scores were averaged after reverse scoring where necessary, with higher scores indicating higher goal commitment. Internal consistency was acceptable at both pre-test and post-test ($\alpha_{pre} = .759$; $\alpha_{post} = .800$).

AI Dependency. AI dependency was measured using the 20-item Generative AI Dependency Scale (Goh et al., 2025). Participants indicated their agreement on a scale from 1

(*Strongly disagree*) to 5 (*Strongly agree*). This scale includes four 5-item subscales measuring (a) Salience (e.g., “I look forward to using generative AI, even if it is unrelated to the task at hand.”), (b) Compulsive Use (e.g., “I find it difficult to reduce or stop using generative AI.”), (c) Negative Consequences (e.g., “I have trouble completing work or other responsibilities without generative AI.”), and (d) Withdrawal (e.g., “I feel unsettled or distracted when I cannot use generative AI.”). Scores were averaged to compute a total dependency with higher scores indicating greater AI dependency. Internal consistency was excellent at both pre-test and post-test ($\alpha_{pre} = .948$; $\alpha_{post} = .961$).

AI Attitudes. AI Attitudes were measured using the 12-item Attitudes Towards Artificial Intelligence Scale (ATTARI-12; Stein et al., 2024). Using a 5-point Likert scale (1= *Strongly disagree*; 5 = *Strongly agree*), participants indicated their agreement with items such as “AI will make this world a better place”. Scores were averaged after reverse-scoring where necessary, with higher scores indicating more positive attitudes towards AI. Internal consistency was acceptable at both pre-test and post-test ($\alpha_{pre} = .882$; $\alpha_{post} = .882$).

AI Trust. Trust in AI was measured with the 8-item XAI Trust Scale (Hoffman et al., 2023). Participants indicated the degree they agree with statements such as “I am confident in the AI. I feel that it works well” on a scale from 1 (*I disagree strongly*) to 5 (*I agree strongly*). Scores were averaged after reverse-scoring a necessary item with higher scores indicating greater trust in AI. Internal consistency was acceptable across pre-test and post-test ($\alpha_{pre} = .749$; $\alpha_{post} = .768$).

Measures of Exploratory Outcomes

Cognitive Failures. Cognitive Failures were measured using the 15-item scale, Cognitive Failures Questionnaire (Broadbent et al., 1982). Participants indicated how frequently each statement reflected their experience over the past 5 days on a scale from 0 (*Never*) to 4 (*Very often*). An example item is, “Do you find you forget why you went from

one part of the house to the other?” Scores were averaged, and higher scores indicated greater cognitive failures. Internal consistency was good across pre-test and post-test ($\alpha_{pre} = .902$; $\alpha_{post} = .914$).

Mind Wandering. Mind wandering was measured using the 5-item Mind Wandering Questionnaire (Mrazek et al., 2013). Participants indicated how well each statement reflected their experience in the past 5 days, on a scale from 1 (*Almost never*) to 6 (*Almost always*). An example item is “I mind-wander during lectures or presentations.” Scores were averaged, with higher scores indicating greater mind wandering. Internal consistency was good across pre-test and post-test ($\alpha_{pre} = .841$; $\alpha_{post} = .864$).

Life Satisfaction. Life satisfaction was measured using a single item (“All things considered, how satisfied are you with your life as a whole”; Cheung & Lucas, 2014). Participants responded on a scale from 0 (*Completely dissatisfied*) to 10 (*Completely satisfied*).

Positive and Negative Affect. State positive and negative affect were assessed using an 18-item scale based on the Circumplex Model of Affect (Barrett & Russell, 1998; Polak et al., 2015). Participants indicated the extent to which they experienced each emotion over the past 5 days on a scale from 1 (*Not at all*) to 5 (*Extremely*). The scale included three positive affect items at each arousal level—high (e.g., energetic), medium (e.g., happy), and low (e.g., calm)—and three negative affect items at each arousal level—high (e.g., angry), medium (e.g., nervous), and low (e.g., dejected). Responses were averaged within each 3-item subscale for each arousal level such that higher scores indicated greater state positive or negative affectivity. Internal consistency was good across pre-test and post-test for positive affect at high ($\alpha_{pre} = .865$; $\alpha_{post} = .904$), medium ($\alpha_{pre} = .868$; $\alpha_{post} = .896$), and low ($\alpha_{pre} = .812$; $\alpha_{post} = .873$) arousal, as well as for negative affect at high ($\alpha_{pre} = .811$; $\alpha_{post} = .800$), medium ($\alpha_{pre} = .859$; $\alpha_{post} = .849$), and low ($\alpha_{pre} = .836$; $\alpha_{post} = .858$) arousal.

Results

Mixed-design analyses of variance were conducted with time (pre-test vs. post-test) as a within-subjects factor and condition (AI vs. Control) as a between-subjects factor ($N = 137$). Significant interactions were followed by simple effect analyses, including paired samples t -tests examining pre-post changes within each condition (0 = Control, 1 = AI) and independent-samples t -tests comparing conditions at each time point. Homogeneity of variance was assessed using Levene's test. In addition to frequentist analyses, Bayesian analyses were conducted to quantify the relative evidence for the null and alternative hypotheses, using JASP interface version 0.19.2. Effect sizes were reported as partial η^2 for main and interaction effects and Cohen's d for pairwise comparisons (Cohen, 1992).

Baseline Comparisons

Baseline differences between the AI and control conditions were examined using independent-samples t -tests for both main and exploratory outcomes. For the main outcomes, no significant baseline differences were observed, with the exception of action planning. Participants in the AI condition reported more plans for executing actions toward their goal at baseline compared to the control group, $t(135) = -2.18, p = .031$. The remaining outcomes did not differ: productivity, $t(135) = 0.89, p = .377$; implementation intent, $t(130.61) = 0.71, p = .482$; goal progress, $t(135) = 0.61, p = .546$; goal realisation, $t(135) = 1.03, p = .306$; coping planning, $t(135) = -0.97, p = .336$; working alliance: $t(135) = -0.38, p = .704$.

Among the exploratory outcomes, no baseline differences were observed between the groups with the exception of goal commitment and AI trust. Participants in the AI condition reported significantly higher goal commitment, $t(135) = -3.14, p = .002$, whereas participants in the control condition reported significantly higher AI trust, $t(135) = 2.06, p = .041$. The other exploratory outcomes showed no significant group differences: AI dependency, $t(135) = 1.47, p = .145$; cognitive failure, $t(135) = -0.46, p = .647$; mind wandering, $t(135) = -0.24,$

$p = .807$; AI attitudes, $t(135) = 0.76, p = .446$; life satisfaction: $t(135) = -0.26, p = .793$; high-arousal positive affect, $t(135) = -0.01, p = .989$; medium-arousal positive affect, $t(135) = 0.24, p = .809$; low-arousal positive affect, $t(135) = 1.17, p = .244$; high-arousal negative affect, $t(135) = -1.03, p = .307$; medium-arousal negative affect, $t(135) = 0.28, p = .778$; low-arousal negative affect, $t(135) = 0.22, p = .828$.

Demographic characteristics were also compared between the two groups.

Independent-samples t -tests indicated no significant differences in age, $t(135) = -0.87, p = .385$ or income, $t(135) = 1.97, p = .051$. Chi-square tests showed no significant differences in sex, $\chi^2(1) = 0.19, p = .665$, or ethnicity, $\chi^2(3) = 1.84, p = .606$. These results suggest that the two conditions were largely comparable at baseline, although some pre-existing differences were observed in action planning, goal commitment, and AI trust.

Main Analyses

Follow-up univariate analyses indicated significant Time $\times$ Condition interactions for productivity, procrastination, and weekly implementation intention, indicating greater improvements in the AI condition group relative to the control groups on these outcomes. No significant Time $\times$ Condition interactions were found for action planning, coping planning, working alliance dimensions, goal progress, or goal realisation (all $p_s \geq .070$; all $BF_{10} = [0.20, 0.83]$), indicating no evidence for the alternative hypothesis and some evidence for the null hypothesis.

Main Goal Pursuit Outcomes

Productivity. A mixed-design ANOVA revealed a significant main effect of time, $F(1, 135) = 18.38, p < .001$, partial $\eta^2 = .120$, indicating that participants reported higher productivity at post-test ($M = 4.04, SD = 0.59$) than at pre-test ($M = 3.80, SD = 0.67$). This increase was larger in the AI condition than in the control condition, as reflected by a significant Time $\times$ Condition interaction, $F(1, 135) = 4.56, p = .035$, partial $\eta^2 = .033$ (Figure

2). Bayesian analysis provided anecdotal evidence favouring the alternative hypothesis for productivity outcomes ($BF_{10} = 1.44$), indicating that the alternative was more likely to be true than was the null model. Follow-up analyses showed that participants in the AI condition showed a significant increase in productivity outcome from pre-test ($M = 3.75$, $SD = 0.71$) to post-test ($M = 4.11$, $SD = 0.58$), $t(70) = 4.62$, $p < .001$, 95% CI [0.21, 0.52], $d = .55$. In contrast, participants in the control condition did not show a significant change, $t(65) = 1.50$, $p = .139$, 95% CI [-0.04, 0.28], $d = .18$.

Procrastination. A mixed-design ANOVA showed a significant main effect of time, $F(1, 135) = 12.58$, $p = .001$, partial $\eta^2 = .085$, indicating that participants reported a decrease in procrastination at post-test ($M = 2.78$, $SD = 0.87$) than at pre-test ($M = 3.03$, $SD = 0.90$). This reduction was larger in the AI condition than in the control condition, as reflected by a significant Time $\times$ Condition interaction, $F(1, 135) = 7.97$, $p = .005$, partial $\eta^2 = .056$ (Figure 2). Bayesian analysis provided moderate evidence favouring the alternative hypothesis ($BF_{10} = 6.56$), suggesting that the alternative model was six times more likely to be true than the null model. Follow-up analyses showed that participants in the AI condition showed a significant decrease in procrastination from pre-test ($M = 3.14$, $SD = 0.97$) to post-test ($M = 2.70$, $SD = 0.91$), $t(70) = -4.27$, $p < .001$, 95% CI [-0.65, -0.24], $d = -.51$. In contrast, participants in the control condition did not show a significant change across time, $t(65) = -0.55$, $p = .584$, 95% CI [-0.23, 0.13], $d = -.07$. The significant interaction indicates that the AI condition was more effective than the control condition in reducing procrastination over time.

Weekly Implementation Intention and Goal Progress. A mixed-design ANOVA revealed a significant main effect of time for weekly implementation intention, $F(1, 135) = 71.38$, $p < .001$, partial $\eta^2 = .346$, indicating that implementation intentions increased from pre-test ($M = 3.41$, $SD = 0.98$) to post-test ($M = 4.18$, $SD = 0.71$). This increase was larger in

the AI condition than in the control condition, as reflected by a significant Time $\times$ Condition interaction, $F(1, 135) = 4.88, p = .029$, partial $\eta^2 = .035$ (Figure 2). Bayesian analyses provided anecdotal evidence in favour of the alternative hypothesis ($BF_{10} = 1.76$).

Specifically, participants in the AI condition showed a significant increase from pre-test ($M = 3.35, SD = 1.10$) to post-test ($M = 4.31, SD = 0.67$) to a large extent, $t(70) = 7.42, p < .001$, 95% CI [0.70, 1.22], $d = .88$. Those in the control condition also improved significantly from the pre-test ($M = 3.47, SD = 0.85$) to post-test ($M = 4.03, SD = 0.72$), $t(65) = 4.51, p < .001$, 95% CI [0.31, 0.81], $d = .56$. Although implementation intention increased significantly in both conditions, the significant interaction indicates that AI condition showed a greater increase than the control condition.

For weekly goal progress, a significant main effect of time was observed, $F(1, 135) = 107.73, p < .001$, partial $\eta^2 = .444$, indicating that perceived goal progress increased from pre-test ($M = 3.23, SD = 1.01$) to post-test ($M = 4.25, SD = 0.71$). However, the Time $\times$ Condition interaction was not significant, $F(1, 135) = 2.72, p = .101$, partial $\eta^2 = .020$, suggesting comparable improvements across the AI and control conditions (Figure 2).

Action and Coping Planning. Mixed-design ANOVAs revealed significant main effects of time for both action planning, $F(1, 135) = 19.29, p < .001$, partial $\eta^2 = .125$, and coping planning, $F(1, 135) = 59.76, p < .001$, partial $\eta^2 = .307$. Scores increased from pre-test (action: $M = 2.80, SD = 0.97$; coping: $M = 2.26, SD = 1.01$) to post-test (action: $M = 3.23, SD = 1.01$; coping: $M = 3.09, SD = 1.06$). However, Time $\times$ Condition interactions were non-significant for action planning, $F(1, 135) = 3.34, p = .070$, partial $\eta^2 = .024$, or coping planning, $F(1, 135) = 0.38, p = .537$, partial $\eta^2 = .003$, suggesting comparable improvements across conditions (Figure 2).

Goal Realisation. A mixed-design ANOVA revealed a significant main effect of time for goal realisation, $F(1, 135) = 72.34, p < .001$, partial $\eta^2 = .349$, indicating that goal

realisation increased from pre-test ($M = 4.12$, $SD = 1.28$) to post-test ($M = 5.28$, $SD = 1.15$).

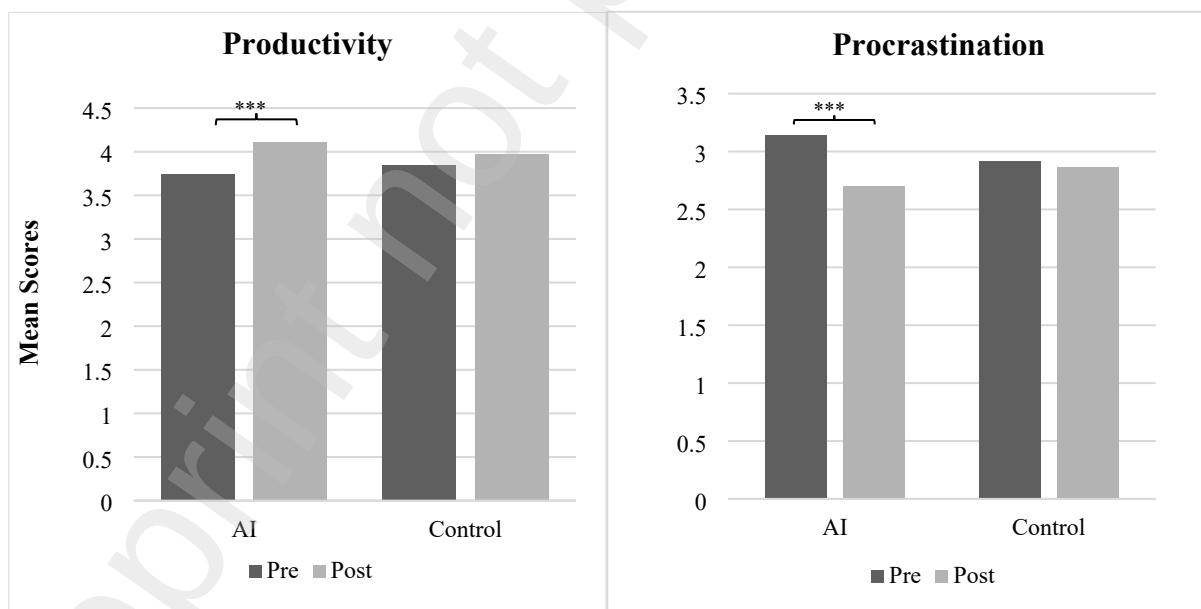
However, the Time $\times$ Condition interaction was not significant, $F(1, 135) = 1.64$, $p = .202$, partial $\eta^2 = .012$, suggesting comparable improvements in both conditions (Figure 3).

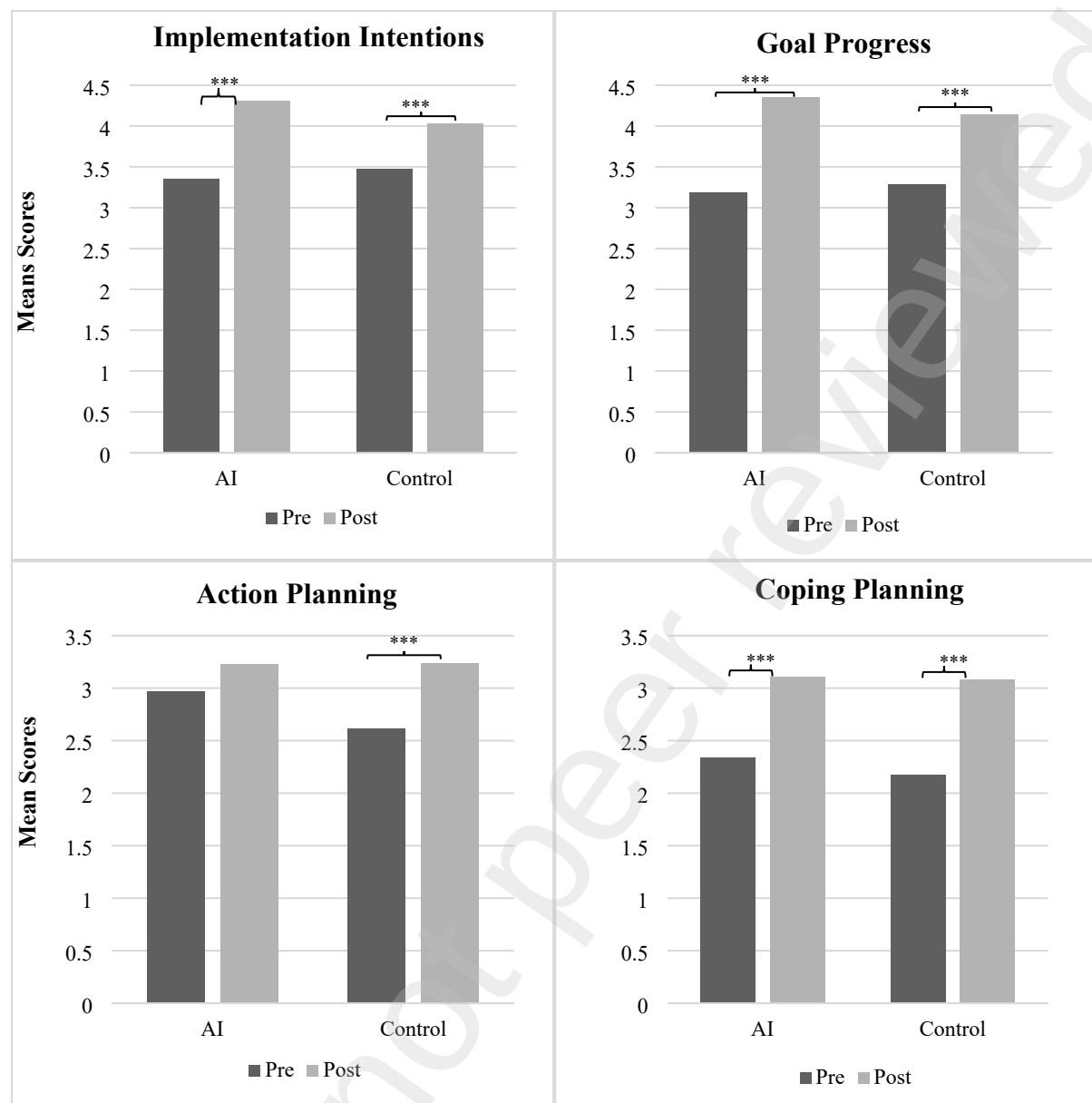
Working Alliance. A mixed-design ANOVA indicated no significant change in working alliance over time, with no significant main effect of time, $F(1, 135) = 0.02$, $p = .892$, partial $\eta^2 = .000$. The Time $\times$ Condition interaction was not significant, $F(1, 135) = 0.19$, $p = .667$, partial $\eta^2 = .001$, suggesting that working alliance remained stable from pre- to post-test across conditions (Figure 3).

Figure 2

Pre- to Post-Test Changes in Main Outcomes Across AI and Control Conditions

(Productivity, Procrastination, Implementation Intentions, Goal Progress, Action Planning, and Coping Planning)

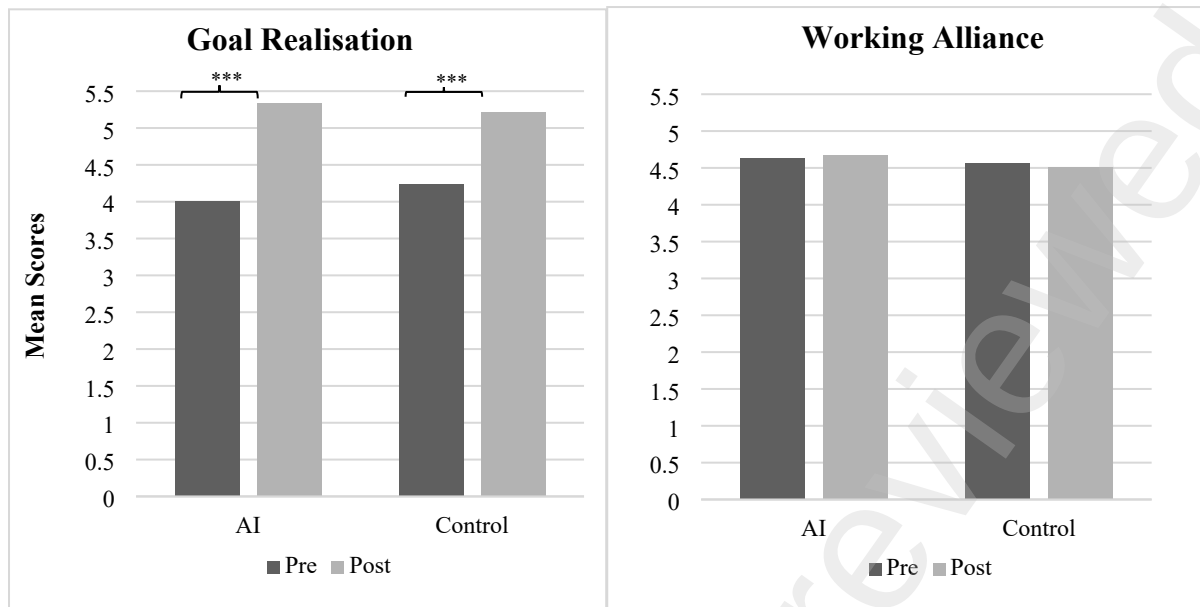




Note. Mean scores are based on a scale range of 1 to 5. The p -values represent the results of paired samples t -tests for the change in scores from pre to post intervention. *** $p < .001$

Figure 3

Pre- to Post-Test Changes in Main Outcomes Across AI and Control Conditions (Goal Realisation, Working Alliance)



Note. Mean scores are based on a scale range of 1 to 7. The p -values represent the results of paired samples t -tests for the change in scores from pre to post intervention. *** $p < .001$

Exploratory Outcomes

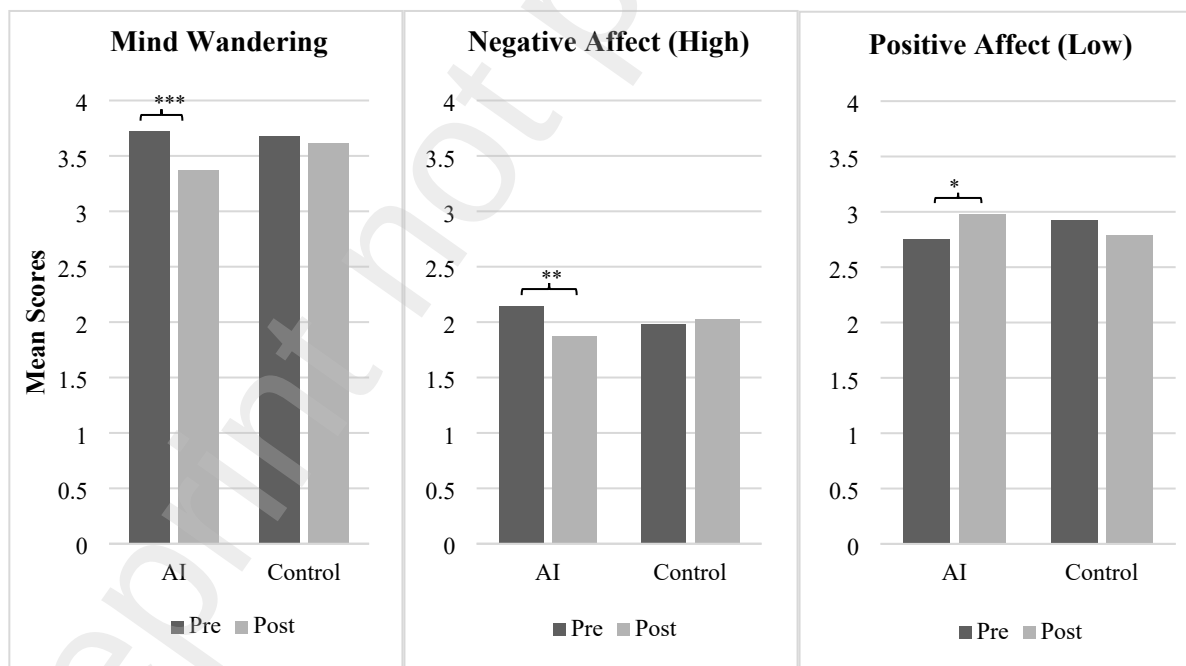
Exploratory analyses were conducted using mixed-design ANOVAs following the same factorial structure as the primary analyses. No significant Time $\times$ Condition interactions were observed for goal commitment, AI dependency, AI attitudes, AI trust, cognitive failures, life satisfaction, high-arousal positive affect, medium-arousal positive affect, medium-arousal negative affect, and low-arousal negative affect (all $ps \geq .094$, $BF_{10} = [0.17, 0.66]$).

Significant Time $\times$ Condition interactions were observed for mind wandering, low-arousal positive affect, and high-arousal negative affect outcomes (Figure 4). Specifically, mind wandering showed a significant main effect of time, $F(1, 135) = 8.73$, $p = .004$, partial $\eta^2 = .061$ and a significant Time $\times$ Condition interaction, $F(1, 135) = 3.91$, $p = .050$, partial $\eta^2 = .028$. Bayesian analysis provided anecdotal evidence favouring the alternative hypothesis ($BF_{10} = 1.07$), indicating a larger reduction in the AI condition from pre-test to post-test ($M_{pre} = 3.72$, $SD_{pre} = 1.15$; $M_{post} = 3.37$, $SD_{post} = 1.01$), whereas no significant change was observed in the control condition.

Low-arousal positive affect showed a significant Time $\times$ Condition interaction, $F(1, 135) = 7.00, p = .009$, partial $\eta^2 = .049$. Bayesian analyses revealed moderate evidence favouring the alternative hypothesis ($BF_{10} = 4.27$). Low-arousal positive affect increased in the AI condition ($M_{pre} = 2.75, SD_{pre} = 0.87; M_{post} = 2.98, SD_{post} = 0.97$), but no significant change was found in the control condition. High-arousal negative affect also showed a significant Time $\times$ Condition interaction, $F(1, 135) = 6.73, p = .011$, partial $\eta^2 = .047$. Bayesian analysis indicated moderate evidence favouring the alternative hypothesis ($BF_{10} = 3.75$). Scores decreased in the AI condition ($M_{pre} = 2.14, SD_{pre} = 0.94; M_{post} = 1.87, SD_{post} = 0.83$), whereas no significant change occurred in the control condition.

Figure 4

Pre- to Post-Test Changes in Exploratory Outcomes Across AI and Control Conditions (Mind Wandering, High-Arousal Negative Affect, Low-Arousal Positive Affect)



Note. Scale range for Mind Wandering is 1 to 6, and for Positive and Negative Affect is 1 to 5. The p -values represent the results of paired samples t -tests for the change in scores from pre to post intervention. *** $p < .001$; ** $p < .01$; * $p < .05$.

Exploratory Moderation Analyses

Exploratory moderation analyses were conducted to examine whether goal commitment, AI dependency, attitudes toward AI, and trust in AI moderated the effects of condition on changes in the main outcomes that showed significant main effects of time. For these analyses, condition was coded as 0 (control) and 1 (AI), and change scores were computed by subtracting pre-test scores from post-test scores. No significant moderation effects were found (all $ps \geq .129$), except AI dependency, which significantly moderated the effect of condition on goal realisation, $B = -0.72$, $SE = 0.35$, $t(133) = -2.03$, $p = .044$.

Table 3*Results of Moderation Analyses*

Main Outcomes	Productivity		Procrastination		Implementation Intention		Goal Progress		Goal Realisation		Action Planning		Coping Planning		Working Alliance	
	<i>B (SE)</i>	<i>p</i>	<i>B(SE)</i>	<i>p</i>	<i>B(SE)</i>	<i>p</i>	<i>B(SE)</i>	<i>p</i>	<i>B(SE)</i>	<i>p</i>	<i>B(SE)</i>	<i>p</i>	<i>B(SE)</i>	<i>p</i>	<i>B(SE)</i>	<i>p</i>
Moderators																
Goal Commitment	0.16 (0.22)	.478	−0.06 (0.27)	.841	0.04 (0.35)	.903	0.25 (0.38)	.522	0.70 (0.53)	.188	−0.24 (0.40)	.541	−0.23 (0.43)	.597	0.24 (0.44)	.588
AI Dependency	−0.10 (0.15)	.488	0.10 (0.18)	.578	−0.04 (0.24)	.868	−0.24 (0.26)	.356	−0.72 (0.35)	.044	−0.40 (0.26)	.129	−0.43 (0.28)	.129	0.13 (0.28)	.647
AI Trust	0.04 (0.21)	.833	−0.22 (0.25)	.382	0.13 (0.33)	.696	0.02 (0.36)	.949	0.20 (0.50)	.684	0.26 (0.37)	.487	0.35 (0.39)	.383	0.07 (0.41)	.865
AI Attitudes	−0.05 (0.18)	.759	0.06 (0.22)	.769	−0.10 (0.28)	.734	−0.31 (0.31)	.321	0.37 (0.43)	.384	0.04 (0.32)	.905	−0.19 (0.34)	.577	0.20 (0.35)	.563

Discussion

The present study provides evidence that daily goal planning through an AI-powered conversational agent yields greater benefits for goal pursuit outcomes than a scripted non-AI chatbot control for three main outcomes. Participants who received coaching from the generative AI chatbot showed significantly greater improvements in productivity and implementation intentions and greater decreases in procrastination. Among the exploratory outcomes, the AI condition also led to a significant increase in low-arousal positive affect and a significant decrease in high-arousal negative affect and mind wandering relative to the control condition. However, several main outcomes improved similarly across both conditions, including goal progress, goal realisation, action and coping planning, and working alliance. These results were observed regardless of participants' baseline characteristics. Exploratory moderation analyses indicated that baseline goal commitment, AI dependency, attitudes toward AI, and trust in AI did not moderate the effect of the condition on any of the main outcomes, with the exception of AI dependency, which moderated the effect of condition on goal realisation. However, baseline AI dependency significantly moderated change in goal realisation, such that the AI condition was beneficial only for participants with low dependency. The implications of these findings are discussed below.

The present findings contribute to theoretical models of self-regulation and goal pursuit by highlighting structured daily goal planning as an effective strategy and suggesting the adaptive AI dialogue may further strengthen key self-regulatory mechanisms. Consistent with self-regulation frameworks, both conditions showed comparable improvements in planning processes (action and coping planning) and goal-related outcomes (goal progress and goal realisation) across the intervention week. This pattern suggests that breaking down a weekly goal into actionable tasks helped narrow the intention-behaviour gap (Carver & Scheier, 1998; Sheeran & Webb, 2016), supporting improved planning and goal-specific

outcomes, regardless of the delivery mode. However, the AI condition offered additional benefits in the enactment of implementation intentions, operationalised as taking concrete steps toward their goal, compared to the control non-AI chatbot. Although participants in both conditions engaged in timeboxing and if-then planning and showed comparable gains in action and coping planning outcomes, the AI condition was more effective in supporting behavioural follow-through. Consistent with implementation intention theory, this benefit may reflect strengthened situation-action linkages (Cohen & Hicks, 2017; Gollwitzer, 1999), supported by an AI chatbot's context awareness in recommending contextually relevant actions and reinforcing accountability toward goal-directed behaviour (Jones et al., 2024). More generally, this effect may reflect qualitatively richer engagement with the AI chatbot, which may prompt future-oriented thinking and sustained action (Dechant et al., 2025).

Beyond supporting implementation intentions, AI support also strengthened broader self-regulatory engagement, as reflected in greater improvements in productivity compared to the control condition. These gains likely reflect increased perceived accomplishment and task completion over the intervention week. The AI chatbot used in the current intervention supported daily goal planning, likely reducing cognitive load associated with routine planning tasks and enabling individuals to allocate cognitive resources to more complex productivity tasks (Iqbal et al., 2025; Marois et al., 2025). Concurrently, participants in the AI condition demonstrated a larger reduction in procrastination, suggesting fewer delays in initiating intended tasks relative to the control condition. These findings suggest that AI coaching can help mitigate self-regulation failure, characterised by delaying intended tasks in favour of short-term temptations (Steel, 2007; Sirois & Pychyl, 2013). As a large language model-driven system, the AI chatbot can break down larger tasks into manageable components and provide actionable next steps, thereby facilitating task initiation and reducing procrastination (Bhattacharjee et al., 2023).

While procrastination reflects a delay in the intended course of action (Steel, 2007; Lay & Silverman, 1996), exploratory analyses indicated that participants in the AI condition experienced greater reductions in mind wandering compared to the control condition. Mind wandering reflects interruptions of task-focused attention by task-unrelated thoughts (Mrazek et al., 2013; Seli et al., 2015), suggesting that the AI chatbot's structured guidance may have supported sustained attentional engagement during goal pursuit. One plausible mechanism is that the AI's decision support, such as predicting relevant next relevant steps and providing concrete suggestions, reduced cognitive load and attenuated attentional drift (Gandhi et al., 2023; Schmidhuber et al., 2021). Importantly, reductions in mind wandering occurred without corresponding changes in general cognitive failures, suggesting that these attentional benefits may occur independently of broader cognitive failures (McVay & Kane, 2010). Overall, these findings suggest that conversational AI may act as a cognitive support system that facilitates sustained attention. These proposed mechanisms remain speculative, as cognitive load and attentional processes were not directly measured in the present study, and should be examined more directly in future research.

In addition to behavioural and attentional outcomes, the AI coaching chatbot also produced affective benefits, including a significant reduction in high-arousal negative affect (i.e., anger, hostile, irritability) and an increase in low-arousal positive affect (i.e., calmness, contentment, relaxation), both occurring along a similar arousal dimension. Reduced high-arousal negative affect may reflect lower friction and frustration arising from uncertainty during goal pursuit, supported by personalised responses and tailored validation by the AI chatbot (Schmidhuber et al., 2021; Hu et al., 2025). Conversely, increases in low-arousal positive affect may reflect greater emotional ease, fostered by encouragement and non-judgemental support that did not impose urgency (Inkster et al., 2018; Terblanche et al., 2022). These results align with previous research suggesting that readily accessible and

responsive AI systems can contribute to well-being benefits and strengthen emotional self-regulation (Alabed et al., 2024; Passmore & Tee, 2023; Sun et al., 2025). These affective improvements suggest that the AI goal planning intervention may have shifted emotional states toward greater calmness, which in turn may support sustained engagement with goal pursuit by promoting focus and reducing stress.

Another key finding of the present study is that, despite the AI chatbot being designed to convey empathy, working alliance remained stable from pre- to post-test, with no increase over time and no differences in change between conditions. Working alliance captured a collaborative partnership with the chatbots across three dimensions—goals, tasks, and bond—reflecting trust in advancing aims, agreement on goal-directed collaborative steps, and a sense of rapport and support, respectively (Bordin, 1979; Goldberg et al., 2022). Consistent with prior work, free-text AI chatbots have been associated with modest increases in bonding, but not with a clear advantage in overall working alliance relative to click-based predefined responses chatbots (Mai et al., 2022). The development of a stronger partnership may require a longer interaction period than the 5-day duration of the present study, as there is limited evidence that users form bonds with AI conversation agents over such brief timeframes (Beatty et al., 2022). Nevertheless, previous research has found that even when a working alliance with the AI chatbot does not develop, most participants can still improve their self-resilience (Ellis-Brush, 2021). The present study observed benefits for certain self-regulatory outcomes, suggesting that individuals can still accept and benefit from AI guidance without a strong working alliance.

The observed gains in goal pursuit and exploratory well-being outcomes highlight the practical potential of AI chatbot-based goal planning as a scalable tool for supporting productivity and everyday well-being across different contexts. For example, such AI elements could be integrated into messaging platforms or existing productivity tools, such as

to-do lists or calendar applications, to support students' self-regulation behaviours.

Supporting the feasibility of this approach, the integration of a semi-generative chatbot into a to-do application led to significant improvements in providing psychological support and stress reduction (Lee et al., 2025). Similar AI-supported interventions may also be applicable in organisational settings, where AI tools could enhance work productivity and performance (Al Naqbi et al., 2024). Beyond productivity enhancements, exploratory findings on well-being have practical implications for the use of AI-assisted coaching in emotionally supportive contexts, extending prior research on AI interventions focused on improving well-being and personal growth (Blyler & Seligman, 2024; Shin, 2025). However, the present intervention was brief and structured relatively straightforward goal planning tasks. More complex coaching domains, such as open-ended counselling or in-depth career coaching, may require further development and ethical safeguards.

Despite the advantages of cost-effectiveness, scalability, and continuous availability of AI chatbot-delivered coaching, the limitations of inaccuracies and ethical concerns regarding confidentiality remain (Ho et al., 2025; Passmore & Tee, 2023; Terblanche, 2022). Considering these limitations and the strengths of human coaches, including nuanced emotional understanding, an AI-human hybrid approach may be promising. In such models, AI-based coaching available between sessions can complement more in-depth human coaching sessions (Bachkirova & Kemp, 2025; Bruning & Boak, 2025). Early ventures are exploring AI-empowered chatbots alongside human coaches (Arakawa & Yakura, 2024; Bojic et al., 2025), suggesting growing interest in, and the potential of, hybrid coaching models that leverage the complementary strengths of AI and human coaches.

Several limitations should be considered when interpreting the present findings. The sample consisted of university students, and the intervention focused on week-long academic or personal goals. These may limit generalisability to broader populations, with different

contextual demands or goal structures (Peterson & Merunka, 2014; Shadish et al., 2002). This relatively homogeneous group may be particularly prone to self-regulatory challenges such as procrastination (Fentaw et al., 2022; Özer, 2011). As digital natives, they may also be more technologically proficient and familiar with AI chatbot-based interventions than other populations (Jones et al., 2010; Singh et al., 2023). Further research should examine the effectiveness of AI-based goal planning interventions in more diverse groups with self-regulation challenges, such as working professionals with high job demands (Bakker & de Vries, 2021), and clinical populations, including individuals with attention-deficit hyperactivity disorder, who may benefit from AI-driven emotional regulation strategies (Faraone et al., 2019; Thakkar et al., 2024).

In addition, the intervention was brief, spanning only 5 days of active use, which constrains conclusions about its longer-term effectiveness. It remains unclear whether the observed benefits would persist after AI support is withdrawn or diminish as novelty fades. Longer intervention designs—spanning multiple weeks or an entire academic term—combined with follow-up assessments, are needed to assess changes in user engagement over time, potential novelty effects, and the extent to which continued interaction supports habit formation and relational processes (Sadeghi et al., 2022). The current study's outcomes relied on self-report measures, which may be susceptible to social desirability bias (Paulhus & Vazire, 2007). Although group assignment was largely successful, pre-existing baseline differences were observed in action planning, goal commitment, and trust in AI, with participants in the AI condition showing a slight initial advantage in action planning. As these variables were not statistically controlled, their influence on outcomes could not be fully ruled out. However, exploratory moderation analyses found that AI attitudes and AI trust did not significantly moderate the effect of condition on all main outcomes, suggesting that the benefits of AI daily goal planning were largely robust across these baseline differences.

Beyond baseline considerations, future research would benefit from incorporating more objective indicators of goal pursuit, such as academic performance or task completion measures, to complement self-report outcomes and strengthen causal inferences (Smyth et al., 2023).

Future research may extend the present work by shifting the AI goal planning intervention from a standalone chat into users' existing digital ecosystems and further exploring relational design elements. Integrating AI coaches with task-management systems or calendars could enable more context-specific prompts, for example, by identifying available time windows (Myers et al., 2007; Rajamoorthy et al., 2025). Such integration would also allow the monitoring of objective indicators, which could support adaptive learning and enable the personalisation of coaching strategies for individual users. Further work may examine user customisation options, including the effectiveness of motivational versus analytical communication styles. While the current intervention adopted a predominantly task-focused design, future studies could explore AI interventions with greater social presence and emotional connection for more emotionally oriented interventions. However, the use of highly anthropomorphic design features may raise ethical concerns, highlighting the importance of ongoing assessment of trust and privacy issues (Osika, 2025; Salles et al., 2020).

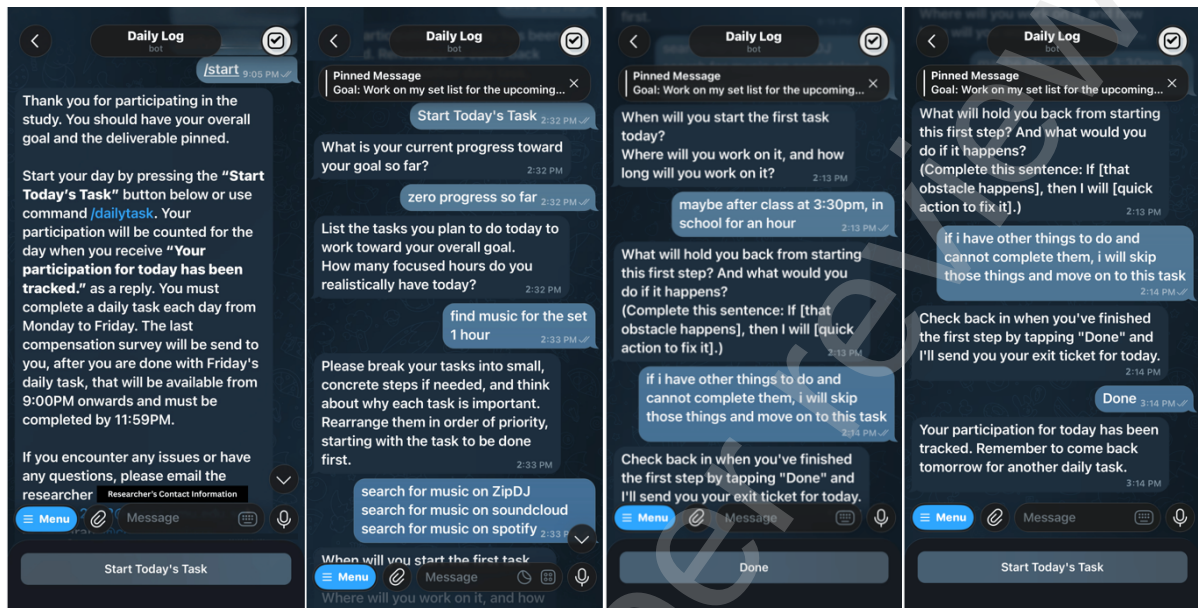
In conclusion, this study highlights the potential of AI-driven daily goal planning as an effective tool for enhancing productivity, reducing procrastination, and improving well-being. Although the intervention was brief, the findings support that the AI-assisted coaching may help address self-regulation challenges. Specifically, the AI chatbot provided personalised guidance that helped students translate their goals into action, follow through on intentions, and manage distractions and emotions more effectively than a rule-based non-AI chatbot. These results contribute to the growing literature on AI coaching by highlighting its

potential as a tool to enhance certain goal pursuit outcomes. Future research should focus on the long-term effectiveness of AI interventions, their integration into existing digital tools, and the development of hybrid approaches that capitalise on the strengths of both AI and human guidance, while remaining mindful of ethical concerns.

Appendix A

Figure A1

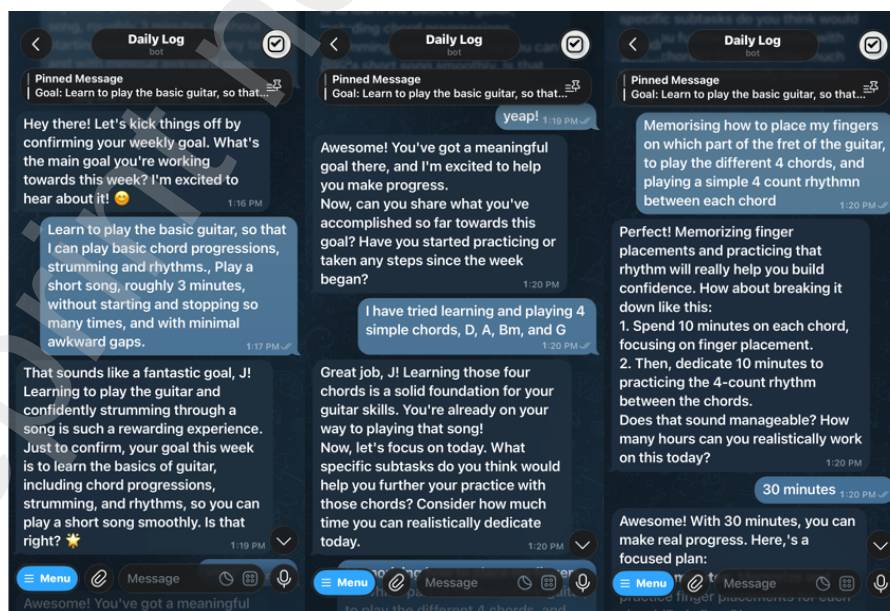
Onboarding and Non-AI Chatbot (Same Script for All Days)

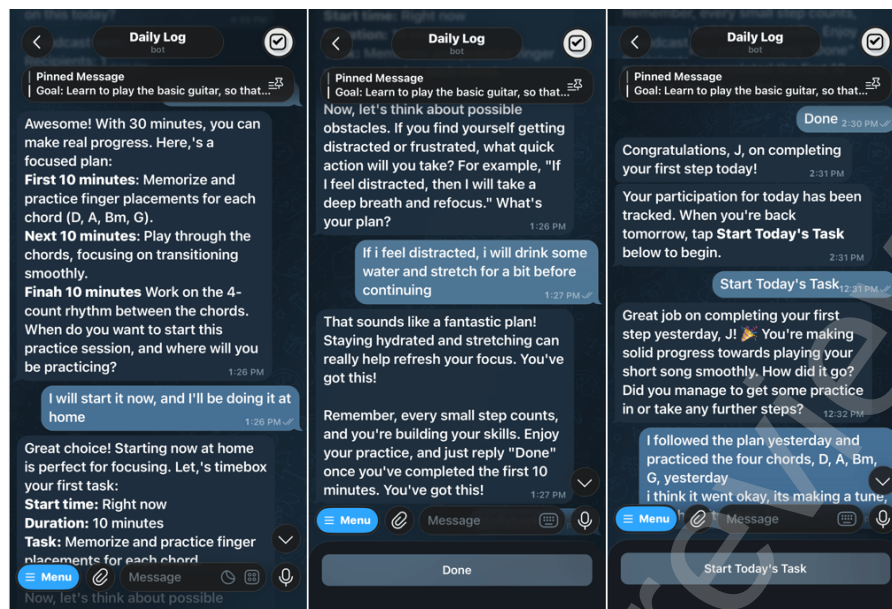


Note. The screenshots shown are simulated responses based on a participant's interaction.

Figure A2

AI Chatbot Adaptive Conversation (Day 1 and Start of Day 2)





Note. The screenshots shown are simulated responses based on a participant's interaction.

References

- Aeon, B., Faber, A., & Panaccio, A. (2021). Does time management work? A meta-analysis. *PloS one*, 16(1), e0245066. <https://doi.org/10.1371/journal.pone.0245066>
- Al Naqbi, H., Bahroun, Z., & Ahmed, V. (2024). Enhancing work productivity through generative Artificial Intelligence: A comprehensive literature review. *Sustainability*, 16(3), 1166. <https://doi.org/10.3390/su16031166>
- Al-Amin, M., Ali, M. S., Salam, A., Khan, A., Ali, A., Ullah, A., Alam, M. N., & Chowdhury, S. K. (2024). History of generative Artificial Intelligence (AI) chatbots: Past, present, and future development. *arXiv*. <https://doi.org/10.48550/arXiv.2402.05122>
- Alabed, A., Javornik, A., Gregory-Smith, D. & Casey, R. (2024). More than just a chat: A taxonomy of consumers' relationships with conversational AI agents and their well-being implications. *European Journal of Marketing*, 58(2), 373–409. <https://doi.org/10.1108/EJM-01-2023-0037>
- Alazraki, L., Ghachem, A., Polydorou, N., Khosmood, F., & Edalat, A. (2021). An empathetic AI coach for self-attachment therapy. *2021 IEEE Third International Conference on Cognitive Machine Intelligence (CogMI)*, 78–87. <https://doi.org/10.1109/CogMI52975.2021.00019>
- Arakawa, R., & Yakura, H. (2024). Coaching copilot: Blended form of an LLM-powered chatbot and a human coach to effectively support self-reflection for leadership growth. *arXiv*. <https://doi.org/10.1145/3640794.3665549>
- Armitage C. J. (2009). Effectiveness of experimenter-provided and self-generated implementation intentions to reduce alcohol consumption in a sample of the general population: A randomized exploratory trial. *Health Psychology*, 28(5), 545–553. <https://doi.org/10.1037/a0015984>

- Bachkirova, T., & Kemp, R. (2025). 'AI coaching': Democratising coaching service or offering an ersatz? *Coaching: An International Journal of Theory, Research and Practice*, 18(1), 27–45. <https://doi.org/10.1080/17521882.2024.2368598>
- Bakker, A. B., & de Vries, J. D. (2021). Job demands–resources theory and self-regulation: new explanations and remedies for job burnout. *Anxiety, Stress, & Coping*, 34(1), 1–21. <https://doi.org/10.1080/10615806.2020.1797695>
- Barrett, L. F., & Russell, J. A. (1998). Independence and bipolarity in the structure of current affect. *Journal of Personality and Social Psychology*, 74(4), 967–984. <https://doi.org/10.1037/0022-3514.74.4.967>
- Beatty, C., Malik, T., Meheli, S., & Sinha, C. (2022). Evaluating the therapeutic alliance with a free-text cbt conversational agent (Wysa): A mixed-methods study. *Frontiers in digital health*, 4, 847991. <https://doi.org/10.3389/fdgth.2022.847991>
- Bhattacharjee, A., Zeng, Y., Xu, S. Y., Kulzhabayeva, D., Ma, M., Kornfield, R., Ahmed, S. I., Mariakakis, A., Czerwinski, M. P., Kuzminykh, A., Liut, M., & Williams, J. J. (2023). Understanding the role of Large Language Models in personalizing and scaffolding strategies to combat academic procrastination. *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, 15, 1–18. <https://doi.org/10.1145/3613904.3642081>
- Billore, S., Anisimova, T., & Vrontis, D. (2023). Self-regulation and goal-directed behavior: A systematic literature review, public policy recommendations, and research agenda. *Journal of Business Research*, 156(113435). <https://doi.org/10.1016/j.jbusres.2022.113435>
- Blyler, A. P., & Seligman, M. E. P. (2024). AI assistance for coaches and therapists. *The Journal of Positive Psychology*, 19(4), 579–591. <https://doi.org/10.1080/17439760.2023.2257642>

Bojic, I., Ong, Q. C., Ito, S., Liu, J., Lawate, A., Palaiyan, M., Nair, E., Lwin, M., Theng, Y.

L., Chia, M., Lee, C. P., Abisheganaden, J., & Car, J. (2025). AI-empowered health coaching for university students: A mixed-method process evaluation. *Computers in Biology and Medicine*, 194, 110271.

<https://doi.org/10.1016/j.compbimed.2025.110271>

Bordin, E.S. (1979). The generalizability of the psychoanalytic concept of the working alliance. *Psychotherapy*, 16, 252-260. <https://doi.org/10.1037/H0085885>

Brenda, K. W. (2024). Humanity's evolving conversations: AI as confidant, coach, and companion. *Cyberpsychology, Behavior and Social Networking*, 27(11), 750–752.

<https://doi.org/10.1089/cyber.2024.0387>

Broadbent, D. E., Cooper, P. F., FitzGerald, P., & Parkes, K. R. (1982). The Cognitive Failures Questionnaire (CFQ) and its correlates. *The British Journal of Clinical Psychology*, 21(1), 1–16. <https://doi.org/10.1111/j.2044-8260.1982.tb01421.x>

Bruning, F., & Boak, G. (2025). Artificial Intelligence coach bots: Coaches' perceptions of potential future impacts on professional coaching, a qualitative study. *Journal of Work-Applied Management*. <https://doi.org/10.1108/JWAM-12-2024-0178>

Burrous, T. C. (2021). *A comparison study on E-coaching and face-to-face coaching* (Doctoral dissertation, University of Pennsylvania). ProQuest Dissertation & Theses Global. (Publication No. 2545535811) <https://eric.ed.gov/?id=ED666124>

Cambron, C., Kosterman, R., Catalano, R. F., Guttmanova, K., Herrenkohl, T. I., Hill, K. G., & Hawkins, J. D. (2017). The role of self-regulation in academic and behavioral paths to a high school diploma. *Journal of Developmental and Life Course Criminology*, 3(3), 304–325. <https://doi.org/10.1007/s40865-017-0066-5>

Carver C. S. (2018). Control theory, goal attainment, and psychopathology. *Psychological inquiry*, 29(3), 139–144. <https://doi.org/10.1080/1047840X.2018.1513681>

- Chawla, N., Gabriel, A. S., Dahling, J. J., & Patel, K. (2016). Feedback dynamics are critical to improving performance management systems. *Industrial and Organizational Psychology: Perspectives on Science and Practice*, 9(2), 260–266.
<https://doi.org/10.1017/iop.2016.8>
- Chernev, A., Böckenholt, U., & Goodman, J. (2015). Choice overload: A conceptual review and meta-analysis. *Journal of Consumer Psychology*, 25(2), 333–358. <https://doi.org/10.1016/j.jcps.2014.08.002>
- Claessens, B. J. C., van Eerde, W., Rutte, C. G., & Roe, R. A. (2007). A review of the time management literature. *Personnel Review*, 36(2), 255–276.
<https://doi.org/10.1108/00483480710726136>
- Cohen J. (1988). *Statistical power analysis for the behavioral sciences*. Routledge.
- Cohen J. (1992). A power primer. *Psychological Bulletin*, 112(1), 155–159.
<https://doi.org/10.1037//0033-2909.112.1.155>
- Cohen, A. L., & Hicks, J. L. (2017). *Prospective memory: Remembering to remember, remembering to forget*. Springer.
- Dechant, M., Lash, E., Shokr, S., & O'Driscoll, C. (2025). Future me, a prospection-based chatbot to promote mental well-being in youth: Two exploratory user experience studies. *JMIR formative research*, 9, e74411. <https://doi.org/10.2196/74411>
- Distefano, R., Grenell, A., Palmer, A. R., Houlihan, K., Masten, A. S., & Carlson, S. M. (2021). Self-regulation as promotive for academic achievement in young children across risk contexts. *Cognitive Development*, 58(101050).
<https://doi.org/10.1016/j.cogdev.2021.101050>
- Ellis-Brush, K. (2021). Augmenting coaching practice through digital methods. *International Journal of Evidence Based Coaching and Mentoring, Spec Iss 15*, 187–197.

- Faraone, S. V., Rostain, A. L., Blader, J., Busch, B., Childress, A. C., Connor, D. F., & Newcorn, J. H. (2019). Practitioner review: Emotional dysregulation in attention-deficit/hyperactivity disorder— implications for clinical recognition and intervention. *Journal of child psychology and psychiatry, and allied disciplines*, 60(2), 133–150. <https://doi.org/10.1111/jcpp.12899>
- Faul, F., Erdfelder, E., Lang, A. G., & Buchner, A. (2007). G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods*, 39, 175–191. <https://doi.org/10.3758/BF03193146>
- Fentaw, Y., Moges, B. & Ismail, S. (2022). Academic procrastination behavior among public university students. *Education Research International*, 5, 1–8. <https://doi.org/10.1155/2022/1277866>
- Fry, J. P., & Neff, R. A. (2009). Periodic prompts and reminders in health promotion and health behavior interventions: systematic review. *Journal of Medical Internet Research*, 11(2), e16. <https://doi.org/10.2196/jmir.1138>
- Gandhi, T. K., Classen, D., Sinsky, C. A., Rhew, D. C., Vande Garde, N., Roberts, A., & Federico, F. (2023). How can artificial intelligence decrease cognitive and work burden for front line practitioners? *JAMIA open*, 6(3), ooad079. <https://doi.org/10.1093/jamiaopen/ooad079>
- Ghisi, G. L., Grace, S. L., Thomas, S., Vieira, A. M., Costa, I. Z., & Oh, P. (2015). Knowledge and exercise behavior maintenance in cardiac rehabilitation patients receiving educational interventions. *Heart & Lung: The Journal of Critical Care*, 44(6), 474–480. <https://doi.org/10.1016/j.hrtlng.2015.09.004>
- Glikson, E., & Woolley, A.W. (2020). Human trust in Artificial Intelligence: Review of empirical research. *Academy of Management Annals*, 14(2), 627–660. <https://doi.org/10.5465/annals.2018.0057>

- Goh, A. Y. H., Hartanto, A., & Majeed, N. M. (2025). Generative artificial intelligence dependency: Scale development, validation, and its motivational, behavioral, and psychological correlates. *Computers in Human Behavior Reports*, 100845. <https://doi.org/10.1016/j.chbr.2025.100845>
- Goldberg, S. B., Baldwin, S. A., Riordan, K. M., Torous, J., Dahl, C. J., Davidson, R. J., & Hirshberg, M. J. (2022). Alliance with an unguided smartphone app: Validation of the digital working alliance inventory. *Assessment*, 29(6), 1331–1345. <https://doi.org/10.1177/10731911211015310>
- Gollwitzer, P. M. (1999). Implementation intentions: Strong effects of simple plans. *American Psychologist*, 54(7), 493–503. <https://doi.org/10.1037/0003-066X.54.7.493>
- Gollwitzer, P. M., & Sheeran, P. (2006). Implementation intentions and goal achievement: A meta-analysis of effects and processes. In M. P. Zanna (Ed.), *Advances in experimental social psychology* (Vol. 38, pp. 69–119). Elsevier Academic Press. [https://doi.org/10.1016/S0065-2601\(06\)38002-1](https://doi.org/10.1016/S0065-2601(06)38002-1)
- Grant, A. M. (2013). The efficacy of coaching. In J. Passmore, D. B. Peterson, & T. Freire (Eds.), *The Wiley-Blackwell handbook of the psychology of coaching and mentoring* (pp. 15–39). Wiley Blackwell.
- Grinschgl, S., Papenmeier, F., & Meyerhoff, H. S. (2021). Consequences of cognitive offloading: Boosting performance but diminishing memory. *Quarterly Journal of Experimental Psychology*, 74(9), 1477–1496. <https://doi.org/10.1177/17470218211008060>
- Hagger, M. S., & Luszczynska, A. (2014). Implementation intention and action planning interventions in health contexts: state of the research and proposals for the way forward. *Applied psychology. Health and well-being*, 6(1), 1–47. <https://doi.org/10.1111/aphw.12017>

- Harkin, B., Webb, T. L., Chang, B. P. I., Prestwich, A., Conner, M., Kellar, I., Benn, Y., & Sheeran, P. (2016). Does monitoring goal progress promote goal attainment? A meta-analysis of the experimental evidence. *Psychological Bulletin*, 142(2), 198–229. <https://doi.org/10.1037/bul0000025>
- Heim, E., Rötger, A., Lorenz, N., & Maercker, A. (2018). Working alliance with an avatar: How far can we go with internet interventions? *Internet interventions*, 11, 41–46. <https://doi.org/10.1016/j.invent.2018.01.005>
- Ho, J. Q., Hu, M., Chen, T. X., & Hartanto, A. (2025). Potential and pitfalls of romantic Artificial Intelligence (AI) companions: A systematic review. *Computers in Human Behavior Reports*, 19, 100715. <https://doi.org/10.1016/j.chbr.2025.100715>
- Hoffman, R., Mueller, S. T., Klein, G., & Litman, J. (2023). Measures for explainable AI: Explanation goodness, user satisfaction, mental models, curiosity, trust, and human-AI performance. *Frontiers Computer Science*, 5. <https://doi.org/10.3389/fcomp.2023.1096257>
- Hollenbeck, J. R., Klein, H. J., O’Leary, A. M., & Wright, P. M. (1989). Investigation of the construct validity of a self-report measure of goal commitment. *Journal of Applied Psychology*, 74(6), 951–956. <https://doi.org/10.1037/0021-9010.74.6.951>
- Hu, M., Chua, X. C. W., Diong, S. F., Kasturiratna, K. T. A. S., Majeed, N. M., & Hartanto, A. (2025). AI as your ally: The effects of AI-assisted venting on negative affect and perceived social support. *Applied psychology. Health and well-being*, 17(1), e12621. <https://doi.org/10.1111/aphw.12621>
- Hung, J. W., Hartanto, A., Goh, A. Y. H., Eun, Z. K. Y., Kasturiratna, K. T. A. S., Lee, Z. X., & Majeed, N. M. (2025). The efficacy of incorporating Artificial Intelligence (AI) chatbots in brief gratitude and self-affirmation interventions: Evidence from two

- exploratory experiments. *Computers in Human Behavior: Artificial Humans*, 4(100151). <https://doi.org/10.1016/j.chbah.2025.100151>.
- Inkster, B., Sarda, S., & Subramanian, V. (2018). An empathy-driven, conversational artificial intelligence agent (Wysa) for digital mental well-being: Real-world data evaluation mixed-methods study. *JMIR mhealth and uHealth*, 6(11), e12106. <https://doi.org/10.2196/12106>
- International Coaching Federation. (2023). *2023 ICF global coaching study: 2023 executive summary*. https://coachingfederation.org/wp-content/uploads/2023/04/2023ICFGlobalCoachingStudy_ExecutiveSummary.pdf
- Iqbal, J., Hashmi, Z. F., Asghar, M. Z., & Abid, M. N. (2025). Generative AI tool use enhances academic achievement in sustainable education through shared metacognition and cognitive offloading among preservice teachers. *Scientific reports*, 15(1), 16610. <https://doi.org/10.1038/s41598-025-01676-x>
- JASP Team. (2024). Introducing JASP 0.19.2: New modules, improved meta-analysis, and much more. <https://jasp-stats.org/2024/12/11/introducing-jasp-0-19-2-new-modules-improved-meta-analysis-and-much-more/>
- Johansson, F., Rozental, A., Edlund, K., Côté, P., Sundberg, T., Onell, C., Rudman, A., & Skillgate, E. (2023). Associations between procrastination and subsequent health outcomes among university students in Sweden. *JAMA Network Open*, 6(1), e2249346. <https://doi.org/10.1001/jamanetworkopen.2022.49346>
- Jones, B., Xu, Y., Li, Q., & Scherer, S. (2024). Designing a proactive context-aware AI chatbot for people's long-term goals. *CHI'24: CHI Conference on Human Factors in Computing Systems*, 1-7. <https://doi.org/10.1145/3613905.3650912>

- Jones, C.R., Ramanau, R., Cross, S., & Healing, G. (2010). Net generation or Digital Natives: Is there a distinct new generation entering university? *Computers & Education*, 54(3), 722-732. <https://doi.org/10.1016/j.compedu.2009.09.022>
- Jones, R. J., Woods, S. A., & Guillaume, Y. R. F. (2016). The effectiveness of workplace coaching: A meta-analysis of learning and performance outcomes from coaching. *Journal of Occupational and Organizational Psychology*, 89(2), 249–277. <https://doi.org/10.1111/joop.12119>
- Khanna, A., Pandey, B., Vashishta, K., Kalia, K., Pradeepkumar, B., & Das, T. (2015). A study of today's A.I. through chatbots and rediscovery of machine intelligence. *International Journal of u- and e- Service, Science and Technology*, 8, 277–284.
- Klingsieck, K. B. (2013). Procrastination: When good things don't come to those who wait. *European Psychologist*, 18(1), 24–34. <https://doi.org/10.1027/1016-9040/a000138>
- Łaguna, M., Mielniczuk, E., & Razmus, W. (2023). Development and initial validation of the Daily Goal Realization Scale. *Current Issues in Personality Psychology*, 11(3), 240–250. <https://doi.org/10.5114/cipp/155945>
- Lay, C., & Silverman, S. (1996). Trait procrastination, anxiety, and dilatory behavior. *Personality and Individual Differences*, 21(1), 61–67. [https://doi.org/10.1016/0191-8869\(96\)00038-4](https://doi.org/10.1016/0191-8869(96)00038-4)
- Lee, S., Jeong, J., Kim, M., Lee, S., Kim, S. P., & Jung, D. (2025). Development of a mobile intervention for procrastination augmented with a semigenerative chatbot for university students: Pilot randomized controlled trial. *JMIR mHealth and uHealth*, 13, e53133. <https://doi.org/10.2196/53133>
- Lin, H., Xu, D., Yang, M., Ma, X., Yan, N., Chen, H., He, S., & Deng, N. (2022). Behaviour change techniques that constitute effective planning interventions to improve physical

- activity and diet behaviour for people with chronic conditions: a systematic review. *BMJ open*, 12(8), e058229. <https://doi.org/10.1136/bmjopen-2021-058229>
- Locke, E. A., & Latham, G. P. (2002). Building a practically useful theory of goal setting and task motivation: A 35-year odyssey. *American Psychologist*, 57(9), 705–717. <https://doi.org/10.1037/0003-066X.57.9.705>
- Luo, B., Lau, R. Y., Li, C., & Si, Y. (2021). A critical review of state-of-the-art chatbot designs and applications. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 12(1), e1434. <https://doi.org/10.1002/widm.1434>
- Mahasneh, A. M., Bataineh, O. T., & Al-Zoubi, Z. H. (2016). The relationship between academic procrastination and parenting styles among Jordanian undergraduate university students. *The Open Psychology Journal*, 9(1), 25–34. <https://doi.org/10.2174/1874350101609010025>
- Mai, V., Neef, C., & Richert, A. (2022). “Clicking vs. Writing”—The impact of a chatbot’s interaction method on the working alliance in AI-based Coaching. *Coaching Theorie & Praxis*, 8, 15–31. <https://doi.org/10.1365/s40896-021-00063-3>
- Marois, A., Lavallée, I., Boily, G., Ramon Alaman, J., Desrosiers, B., & Lavoie, N. (2025). Chatbot memory: Uncovering how mental effort and chatbot interactions affect short-term learning. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 69(1), 2120–2126. <https://doi.org/10.1177/10711813251358242>
- McVay, J. C., & Kane, M. J. (2010). Does mind wandering reflect executive function or executive failure? Comment on Smallwood and Schooler (2006) and Watkins (2008). *Psychological Bulletin*, 136(2), 188–197. <https://doi.org/10.1037/a0018298>
- Miller, G. E., Yu, T., Chen, E., & Brody, G. H. (2015). Self-control forecasts better psychosocial outcomes but faster epigenetic aging in low-SES youth. *Proceedings of*

- the National Academy of Sciences of the United States of America*, 112(33), 10325–10330. <https://doi.org/10.1073/pnas.1505063112>
- Mrazek, M. D., Phillips, D. T., Franklin, M. S., Broadway, J. M., & Schooler, J. W. (2013). *Mind Wandering Questionnaire (MWQ)* [Database record]. APA PsycTests. <https://doi.org/10.1037/t36473-000>
- Myers, K. L., Berry, P. M., Blythe, J., Conley, K., Gervasio, M. T., McGuinness, D. L., Morley, D. N., Pfeffer, A., Pollack, M. E., & Tambe, M. (2007). An Intelligent Personal Assistant for Task and Time Management. *The AI Magazine*, 28, 47–61. <https://doi.org/10.1609/aimag.v28i2.2039>
- Noar, S. M., Grant Harrington, N., Van Stee, S. K., & Shemanski Aldrich, R. (2011). Tailored health communication to change lifestyle behaviors. *American Journal of Lifestyle Medicine*, 5(2), 112–122. <https://doi.org/10.1177/1559827610387255>
- OpenAI. (2024, July 18). *GPT-4o mini: Advancing cost-efficient intelligence*. <https://openai.com/index/gpt-4o-mini-advancing-cost-efficient-intelligence/>
- Osika, G. (2025). Anthropomorphic design: ethical dilemmas. *Scientific Papers of Silesian University of Technology Organization and Management Series*, 233, 365–383. <https://doi.org/10.29119/1641-3466.2025.233.25>
- Özer, B.U. (2011). A cross sectional study on procrastination: who procrastinate more? 2011 *International Conference on Education, Research and Innovation IPEDR*, 18. <https://api.semanticscholar.org/CorpusID:43422017>
- Passmore, J., & Tee, D. (2023). Can Chatbots replace human coaches? Issues and dilemmas for the coaching profession, coaching clients and for organisations. *The Coaching Psychologist*, 19(1), 47–54. <https://doi.org/10.53841/bpstep.2023.19.1.47>

- Paulhus, D. L., & Vazire, S. (2007). The self-report method. In R. W. Robins, R. C. Fraley, & R. F. Krueger (Eds.), *Handbook of research methods in personality psychology* (pp. 224–239). The Guilford Press.
- Peterson, R. A., & Merunka, D. R. (2014). Convenience samples of college students and research reproducibility. *Journal of Business Research*, 67, 1035-1041.
- Pirolli, P., Mohan, S., Venkatakrishnan, A., Nelson, L., Silva, M., & Springer, A. (2017). Implementation intention and reminder effects on behavior change in a mobile health system: A predictive cognitive model. *Journal of Medical Internet Research*, 19(11), e397. <https://doi.org/10.2196/jmir.8217>
- Polak, M. A., Richardson, A. C., Flett, J. A. M., Brookie, K. L., & Conner, T. S. (2015). Measuring mood: Considerations and innovations for nutrition science. In L. Dye, and T. Best (Eds.) *Nutrition for Brain Health and Cognitive Performance* (pp. 93–119). London, UK: Taylor and Francis
- Rajamoorthy, R., Karthik, S., Savitha, S., Sindhu, M., Devi, S.V., & Vihitha, V. (2026). Enhancing productivity in educational institutions with an AI-powered real-time schedule management system. In V. S. Rathore, J. M. R. S. Tavares, M. Hanumanthappa, & B. Surendiran (Eds.), *Universal Threats in Expert Applications and Solutions* (pp. 497-511). Springer. https://doi.org/10.1007/978-981-96-7289-9_40
- Riddell, H., Sedikides, C., Gucciardi, D. F., Ben, J., Thøgersen-Ntoumani, C., & Ntoumanis, N. (2022). Goal motives and mental contrasting with implementation intentions facilitate strategic goal persistence and disengagement. *Journal of Applied Social Psychology*, 52(11), 1094–1116. <https://doi.org/10.1111/jasp.12915>
- Rozental, A., Forsström, D., Hussoon, A., & Klingsieck, K. B. (2022). Procrastination among university students: Differentiating severe cases in need of support from less severe

- cases. *Frontiers in psychology*, 13, 783570.
<https://doi.org/10.3389/fpsyg.2022.783570>
- Sadeghi, S., Gupta, S., Gramatovici, S., Lu, J., Ai, H., & Zhang, R. (2022). Novelty and primacy: A long-term estimator for online experiments. *Technometrics*, 64(4), 524–534. <https://doi.org/10.1080/00401706.2022.2124309>
- Salles, A., Evers, K., & Farisco, M. (2020). Anthropomorphism in AI. *AJOB neuroscience*, 11(2), 88–95. <https://doi.org/10.1080/21507740.2020.1740350>
- Schmidhuber, J., Schlögl, S., & Ploder, C. (2021). Cognitive load and productivity implications in human-chatbot interaction. *2021 IEEE 2nd International Conference on Human-Machine Systems (ICHMS)*, 1–6.
<https://doi.org/10.1109/ICHMS53169.2021.9582445>
- Seli, P., Cheyne, J. A., Xu, M., Purdon, C. L., & Smilek, D. (2015). Motivation, intentionality, and mind wandering: Implications for assessments of task-unrelated thought. *Journal of Experimental Psychology. Learning, memory, and cognition*, 41(5), 1417–1425. <https://doi.org/10.1037/xlm0000116>
- Shadish, W. R., Cook, T. D., & Campbell, D. T. (2002). *Experimental and quasi-experimental designs for generalized causal inference*. Boston: Houghton Mifflin.
- Sheeran, P. (2002). Intention—Behavior relations: A conceptual and empirical review. *European Review of Social Psychology*, 12(1), 1–36.
<https://doi.org/10.1080/14792772143000003>
- Sheeran, P., & Webb, T. L. (2016). The intention–behavior gap. *Social and Personality Psychology Compass*, 10(9), 503–518. <https://doi.org/10.1111/spc3.12265>
- Sheeran, P., Webb, T. L., & Gollwitzer, P. M. (2006). Implementation intentions: Strategic automatization of goal striving. In D. de Ridder, & J. de Wit (Eds.), *Self-regulation in Health Behavior* (pp. 119–145). Wiley.

- Shin, Y. (2025). Toward human-centered Artificial Intelligence for users' digital well-being: Systematic review, synthesis, and future directions. *JMIR Human Factors*, 12, e69533. <https://doi.org/10.2196/69533>
- Singh, H., Tayarani-Najaran, M.-H., & Yaqoob, M. (2023). Exploring computer science students' perception of ChatGPT in higher education: A descriptive and correlation study. *Education Sciences*, 13(9), 924. <https://doi.org/10.3390/educsci13090924>
- Sirois, F. M., & Pychyl, T. A. (Eds.). (2016). *Procrastination, health, and well-being*. Elsevier Academic Press.
- Sirois, F., & Pychyl, T. (2013). Procrastination and the priority of short-term mood regulation: Consequences for future self. *Social and Personality Psychology Compass*, 7(2), 115–127. <https://doi.org/10.1111/spc3.12011>
- Smyth, A., Milyavskaya, M., Friese, M., Werner, K., Frech, M.-L., Loschelder, D., Anderson, J., Inzlicht, M., Kolbuszewska, M., & Wang, K. (2023). What constitutes successful goal pursuit? Exploring the relation between subjective and objective measures of goal progress. *Personality Science*, 4(1). <https://doi.org/10.5964/ps.12017>
- Sniehotta, F. F., Schwarzer, R., Scholz, U., & Schüz, B. (2005). Action planning and coping planning for long-term lifestyle change: Theory and assessment. *European Journal of Social Psychology*, 35(4), 565–576. <https://doi.org/10.1002/ejsp.258>
- Steel, P. (2007). The nature of procrastination: A meta-analytic and theoretical review of quintessential self-regulatory failure. *Psychological Bulletin*, 133(1), 65–94. <https://doi.org/10.1037/0033-2909.133.1.65>
- Steel, P. D. G. (2002). *The measurement and nature of procrastination* (Doctoral dissertation). Dissertation Abstracts International: Section B: The Sciences and Engineering, 63(3-B), 1599.

- Steel, P., & Klingsieck, K. B. (2016). Academic procrastination: Psychological antecedents revisited. *Australian Psychologist*, 51(1), 36–46. <https://doi.org/10.1111/ap.12173>
- Stein, J. P., Messingschlager, T., Gnambs, T., Hutmacher, F., & Appel, M. (2024) Attitudes towards AI: Measurement and associations with personality. *Scientific Reports*, 14(2909). <https://doi.org/10.1038/s41598-024-53335-2>
- Sun, X., Jahangir, A., Alghamdi, A. A., & Liu, F. (2025). Use of AI-based mental health tools and psychological well-being among Chinese university students: A parallel mediation model of emotional self-efficacy and perceived autonomy. *Scientific Reports*, 15(1), 40305. <https://doi.org/10.1038/s41598-025-24013-8>
- Terblanche, N., Modyn, J., de Haan, E., & Nilsson, V. O. (2022). Comparing Artificial Intelligence and human coaching goal attainment efficacy. *PloS one*, 17(6), e0270255. <https://doi.org/10.1371/journal.pone.0270255>
- Thakkar, A., Gupta, A., & De Sousa, A. (2024). Artificial Intelligence in positive mental health: A narrative review. *Frontiers in Digital Health*, 6, 1280235. <https://doi.org/10.3389/fdgth.2024.1280235>
- Theeboom, T., Beersma, B., & van Vianen, A. E. M. (2014). Does coaching work? A meta-analysis on the effects of coaching on individual level outcomes in an organizational context. *The Journal of Positive Psychology*, 9(1), 1–18. <https://doi.org/10.1080/17439760.2013.837499>
- Thürmer, J. L., Scheier, M. F., & Carver, C. S. (2020). On the mechanics of goal striving: Experimental evidence of coasting and shifting. *Motivation Science*, 6(3), 266–274. <https://doi.org/10.1037/mot0000157>
- Uhlig, L., Baumgartner, V., Prem, R., Siestrup, K., Korunka, C., & Kubicek, B. (2023). A field experiment on the effects of weekly planning behaviour on work engagement,

- unfinished tasks, rumination, and cognitive flexibility. *Journal of Occupational and Organizational Psychology*, 96(3), 575–598. <https://doi.org/10.1111/joop.12430>
- Van Eerde, W. (2000). Procrastination: Self-regulation in initiating aversive goals. *Applied Psychology: An International Review*, 49(3), 372–389. <https://doi.org/10.1111/1464-0597.00021>
- Wieber, F., Thürmer, J. L., & Gollwitzer, P. M. (2015). Promoting the translation of intentions into action by implementation intentions: Behavioral effects and physiological correlates. *Frontiers in human neuroscience*, 9, 395. <https://doi.org/10.3389/fnhum.2015.00395>
- Yigci, D., Eryilmaz, M., Yetisen, A. K., Tasoglu, S., & Ozcan, A. (2025). Large language model-based chatbots in higher education. *Advanced Intelligent Systems*, 7(3), 2400429. <https://doi.org/10.1002/aisy.202400429>